

MARTIN KOOME

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PROFILE

An AI Engineer and Master's student in Engineering Artificial Intelligence at Carnegie Mellon University Africa, specializing in machine learning, deep learning, and probabilistic modeling. I build and deploy end-to-end AI systems across healthcare, education, and enterprise domains, with experience in computer vision, predictive modeling, and edge AI applications. My work combines strong mathematical foundations with practical implementation using PyTorch and modern ML frameworks to deliver scalable, production-ready solutions. I also design cloud and containerized systems using Docker and related tooling to support reliable deployment and automation. I am focused on translating complex data into impactful, real-world systems that improve efficiency, accessibility, and decision-making.

EDUCATION

Masters of Science in Engineering AI

August 2025 - May 2027

Carnegie Mellon University, Africa

Concentration: Probabilistic Graphical Models and Applied Machine Learning.

Achievement: Community Spirit Award for outstanding collaboration and leadership during induction.

Bsc Mathematics and Computer Science (Statistics Major)

September 2021 - May 2025

Multimedia University of Kenya, Nairobi, Kenya

Coursework: Measure and Probability, Operations Research, Time Series Analysis, Stochastic Processes, Survival Analysis.

Project: Survival Analysis of Student Dropout in STEM.

Degree Award: Graduated with First Class Honors.

EXPERIENCE

AI Engineer

May 2024 - Present

Zoner Health, Kigali, Rwanda

- Developed and maintained core features for the Zoner Pharmacy Management System, streamlining pharmacy inventory management and helping reduce stock discrepancies and insurance claim rejections.
- Implemented ZATA-compliant e-invoicing integration, enabling automated tax reporting and supporting regulatory compliance for pharmacies using the platform.
- Conducted user interviews and gathered product feedback from pharmacists and stakeholders, translating real-world operational challenges into software improvements that enhanced usability and customer experience.

Student Success Coordinator

January 2024 - April 2025

Carnegie Mellon University, Kigali, Rwanda

- Collaborated within a six-member interdisciplinary team to curate and organize 300+ academic resources, creating a centralized knowledge base that improved access to supplementary learning materials for students and faculty.
- Designed, developed, and deployed a digital Resource Hub featuring structured learning pathways, resource cataloging, and role-based access management, using Docker containers, nginx, and a virtual machine to deliver a scalable, production-ready platform for multiple stakeholder groups.
- Planned and facilitated technical workshops and interactive training sessions for students and teaching assistants, translating complex technical concepts into practical learning experiences while incorporating user feedback to continuously improve educational resources.

Technical Researcher

September 2024 - August 2025

GoJumpStart, Nairobi, Kenya

- Designed and tested business automation workflows using n8n, aligning solutions to client requirements and improving operational efficiency.
- Configured and optimized server environments, including website hosting with NPM and Portainer, resulting in stable uptime and streamlined resource allocation.
- Developed a Minimum Viable Product for a gym automation system using ERPNext and Frappe, enabling the digitization of membership, payments, and scheduling processes while reducing reliance on manual paper-based records.

PROJECTS

ImaraCare | Harvard Health Systems Innovation Lab Venture Program

February 2024 - June 2025

- Developed a contactless neonatal monitoring system using Wi-Fi Channel State Information and deep learning to estimate heart rate and respiratory rate, targeting low-cost continuous monitoring in resource-constrained neonatal units.
- Processed and engineered CSI signal data by adapting the Pulse-Fi pipeline, identifying neonatal-relevant frequency bands for physiological signal extraction and improving signal quality for downstream learning models.
- Implemented an end-to-end edge AI prototype integrating Wi-Fi sensing, signal processing, and on-device inference using ESP32 microcontrollers, enabling real-time monitoring capabilities in a low-power embedded environment.
- Collaborated within a multidisciplinary team across AI, embedded systems, and healthcare domains to build and validate the system, contributing to a prototype that achieved Top 20 global ranking among 1,553 teams in the Harvard Health Systems Innovation Lab Venture Building Program.

Campus Link | Industry Innovation Lab, CMU Africa

October 2025 - Present

- Led the development of Campus Link, a student-centered platform designed to streamline access to academic and administrative services at CMU Africa.
- Conducted user needs assessments through interviews with students and administrative staff to identify service access bottlenecks.
- Designed and implemented retrieval-augmented generation pipelines using LightRAG, evaluating and selecting language models best suited to the platform's use case.
- Currently working as a student success administration support by aligning the product functionality with both student needs and institutional workflows among other duties that ensure student success.
- Utilized tools such as Python (PyTorch, Transformers), TypeScript, Hugging Face, Git, light RAG and Docker.

SKILLS

- **Technical:** Advanced skills in Python, Machine Learning, PyTorch, TensorFlow, Git version control , n8n automations, ERP Next integrations and intermediate skills in Docker and Cloud Operations.
- **Communication:** Strong written and verbal communication skills, honed through previous leadership and volunteer roles.
- **Leadership & Teamwork:** Managed to lead projects, organize tasks and maintain clear communication that strengthen team performance.
- **Languages:** Fluent in both English and Swahili.

INTERESTS & ACHIEVEMENTS

- **Data Science Challenges:** Ranked as the top 7th Kenyan on Zindi, the largest platform of data scientists and AI builders.
- **Mentorship:** Volunteered as data science co-lead for more than two years to mentor and provide training to fellow students, and was elected as a delegate in the final year.
- **Leadership:** Serving as Vice President of Public Relations for the Toastmasters Club, leading communication strategies, promoting club activities, and engaging members through outreach and visibility initiatives.

REFEREE

Name: Njodzeka Ulrich Ndzerem.

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